SCOPE-IA 4.0: A Multicriteria Predictive Optimization Framework for Intelligent Supply Chain Management

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Abstract: The increasing complexity of global supply chains demands predictive, adaptive, and optimization-oriented decision systems. This paper introduces SCOPE-IA 4.0 (Systemic Control and Optimization for Predictive Enterprises – Artificial Intelligence 4.0), a formal mathematical framework that integrates predictive analytics, multicriteria evaluation, and heuristic optimization for Supply Chain 4.0 environments. The system formalizes five steps of the supply chain (Planning, Procurement, Production, Delivery, and Reverse Logistics), using a compensatory linear scoring function combined with adaptive weights and genetic algorithms. The implementation is carried out in a web environment, with HTML5, CSS3, JavaScript ES6 and open-source libraries such as Chart.js, jsPDF, SheetJS and MathJax. Preliminary results indicate a reduction in operational uncertainty, improved overall efficiency and support for data-driven strategic decision-making. The project is academic, non-profit, and available as a learning and development resource on GitHub.

Keywords: Supply Chain 4.0; Multicriteria Decision; Predictive Analytics; Genetic Algorithms; AI optimization.

1. Introduction

Modern supply chains are characterized by volatility, uncertainty, complexity, and ambiguity (VUCA), requiring managers to adopt predictive and adaptive strategies to maintain resilience and operational efficiency. Traditional management systems rely on historical metrics and reactive adjustments, limiting the ability to anticipate and optimize (Lambert, Cooper, & Pagh, 1998).

Industry 4.0 offers opportunities for real-time data integration, predictive modeling, and optimization algorithms, enabling supply chains to evolve from reactive models to intelligent, adaptive models (Christopher, 2016; Choi, Chan & Yue, 2019). In this context, **SCOPE-IA 4.0** proposes a formal framework that combines multicriteria assessment,

bottleneck simulation, and heuristic optimization, supporting strategic and operational decisions in a replicable and auditable manner.

This research proposes SCOPE-IA 4.0, a formalized predictive optimization framework designed to:

- Normalize operational indicators
- Evaluate multicriteria performance
- Simulate bottlenecks
- Apply heuristic optimization
- Generate executive dashboards for decision making

The research question that guides this study is: How can a mathematically formalized and multicriteria framework, integrated with predictive algorithms, improve systemic efficiency in Supply Chain 4.0 environments?

2. Theoretical Framework

The literature on Supply Chain Management (SCM) emphasizes the importance of operational integration to reduce inefficiencies and increase resilience (Lambert, Cooper, & Pagh, 1998). Christopher (2016) points out that visibility and prediction of events are fundamental for strategic decision-making. Sheffi (2005) argues that organizational resilience depends on the ability to anticipate bottlenecks and proactively adjust operations.

The application of Artificial Intelligence in SCM enables demand forecasting, predictive maintenance, route optimization, supplier risk assessment, and reverse logistics classification (Choi, Chan, & Yue, 2019). However, many existing systems operate as isolated modules, without formal multicriteria integration. SCOPE-IA 4.0 fits into this context by unifying:

- Multicriteria decision theory
- Linear compensatory scoring
- Standardization of indicators
- Genetic heuristic optimization
- Predictive bottleneck simulation

This approach ensures scientific rigor, interpretability, and alignment with the demands of Supply Chain 4.0. Therefore, SCOPE-IA 4.0 formalizes this academic argument

by proposing a multicriteria mathematical framework that integrates indicator normalization, bottleneck simulation, and heuristic optimization. Formalization is supported by the literature of multicriteria decision and genetic heuristics, ensuring scientific rigor.

3. Project SCOPE-IA 4.0

Supply Chain Management **(SCM)** involves interconnected macro-processes, including:

- **CRM** (Customer Relationship Management),
- **ISCM** (Integrated Supply Chain Management) and
- **SRM** (Supplier Relationship Management).

According to the fundamental theory of SCM, operational integration reduces inefficiencies and increases resilience**.**

Applications of Artificial Intelligence in SCM typically include:

- Demand forecasting
- Predictive maintenance
- Route optimization
- Supplier risk assessment
- Classification in reverse logistics

However, existing implementations often function as *fragmented modules*, rather than integrated systems optimization frameworks.

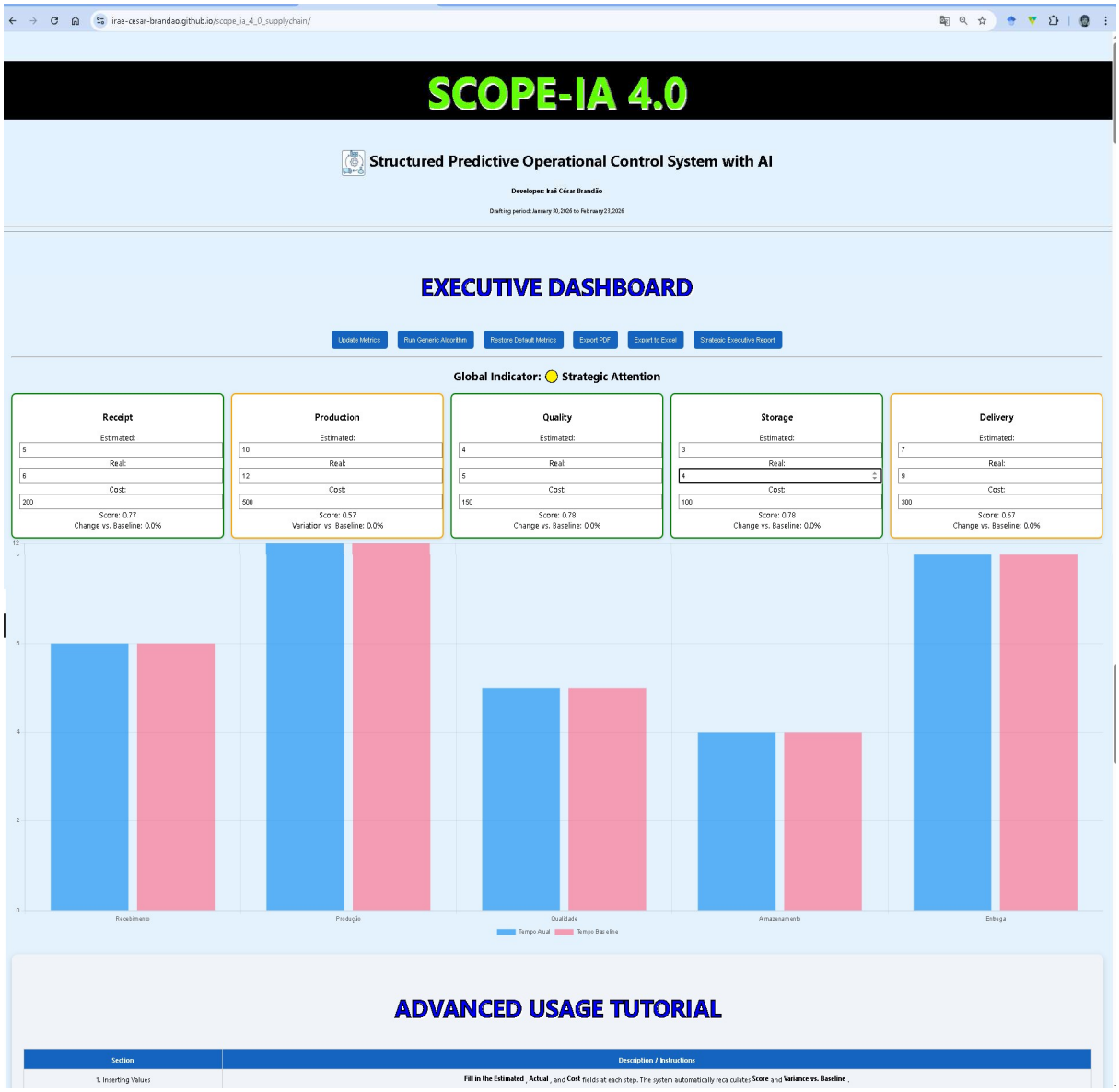
SCOPE-IA 4.0 fills this gap by unifying**:**

- Multicriteria decision theory
- Linear compensatory scoring
- Normalization methods
- Genetic heuristic optimization
- Predictive simulation

The author's GitHub repository makes the Framework available for public access, as per the following link:

SCOPE-IA 4.0 <https://irae-cesar-brandao.github.io/scope_ia_4_0_supplychain/>

The framework and its partial representation can be seen as shown in the following figure:



Prepared by the author (BRANDÃO, 2026).

Figure 1: SCOPE-IA 4.0 Framework Prototype

3.1. Framework Structure

a) Formal Mathematical Structure

The system is defined as:

$$\boldsymbol{S} = (\boldsymbol{E}, \boldsymbol{P}, \boldsymbol{W}, \boldsymbol{F})$$

Where:

- $\boldsymbol{E} = \{\boldsymbol{e_1}, \boldsymbol{e_2}, \boldsymbol{e_3}, \boldsymbol{e_4}, \boldsymbol{e_5}\}$

  Representing: Planning, Procurement, Production, Delivery and Reverse Logistics.
- $\boldsymbol{P} = \{\boldsymbol{T^{estimado}}, \boldsymbol{T^{real}}, \boldsymbol{C}\}$

  operating parameters.
- $\boldsymbol{W} = \{\boldsymbol{\alpha}, \boldsymbol{\beta}\}$

  Vector of adaptive weights.
- $\boldsymbol{F}$

  Global objective function.

b) Performance Function by Step

For each stage:i

$$\boldsymbol{S_i} = \boldsymbol{\alpha}\left(\frac{\boldsymbol{T_i^{estimado}}}{\boldsymbol{T_i^{real}}}\right) + \boldsymbol{\beta}\left(\mathbf{1} - \frac{\boldsymbol{C_i}}{\boldsymbol{C_{max}}}\right)$$

Subject to:

$$\boldsymbol{T_i^{real}} > \mathbf{0}$$

c) Global Optimization Function

$$\mathbf{max}\ \boldsymbol{F} = \sum_{\boldsymbol{i}=\mathbf{1}}^{\mathbf{5}} \boldsymbol{S_i}$$

d) Genetic Optimization Module

The optimization process applies a simplified genetic heuristic**:**

- Initialize the Weight Vector Population
- Assessing fitness through FFF
- Select the best individuals
- Crossover and mutation
- Iterate to convergence

Fitness Function:

$$\boldsymbol{Fitness}(\boldsymbol{W}) = \boldsymbol{F}(\boldsymbol{W})$$

e) Bottleneck Simulation

A bottleneck condition is defined when:

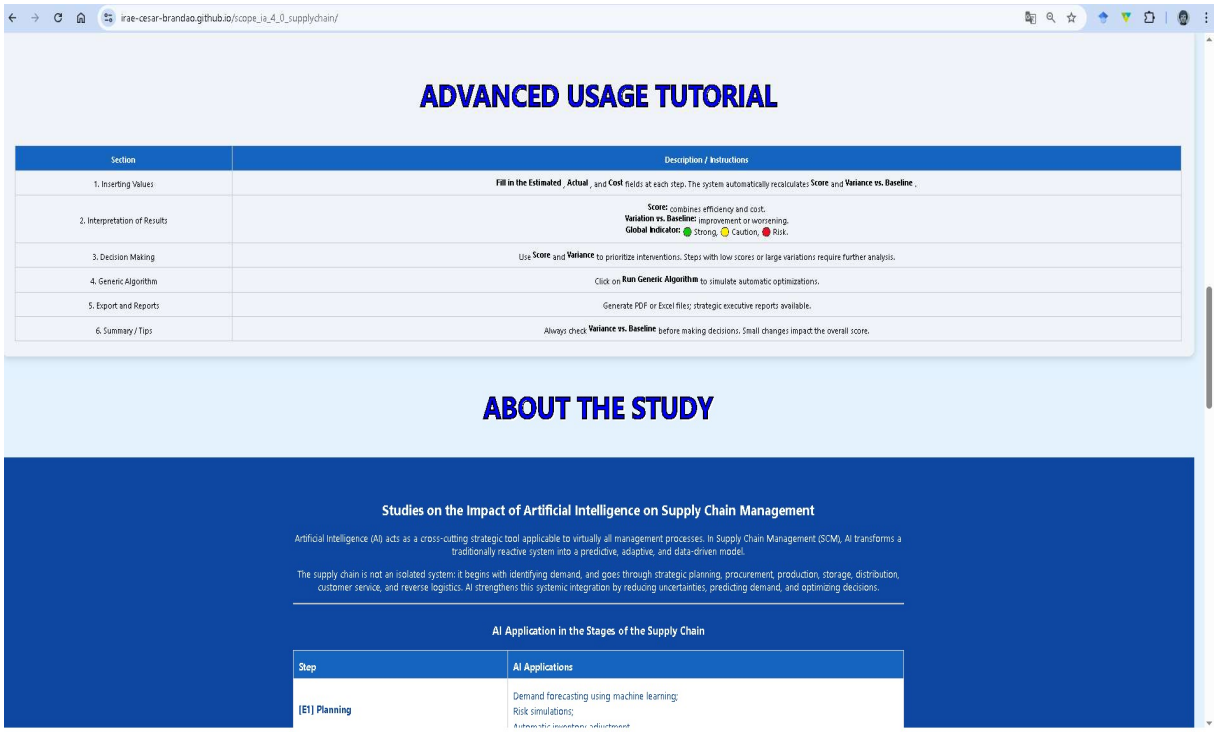
$$\frac{T_i^{real}}{T_i^{estimado}} > \delta$$

Where is a threshold parameter.δ

f) Functional and Non-Functional Resources

- **Functional (RF) features:** interactive dashboards, bottleneck simulation, executive reporting, data export, and adaptive optimization.
- **Non-functional features (RFN):** interoperability, portability, scalability, replicability, and academic transparency.
- **Limitations:** Reliance on reliable data, compensatory linear assumptions, and the need for empirical validation in real corporate environments.

All functionality guidance, manipulation and obtaining of results and exports, are distributed in the framework's browser, as shown in figure 2.



Prepared by the author (BRANDÃO, 2026).

Figure 2: Prototype Framework: Use and Study Tutorial

4. Methodology

The project methodology included:

i. **Mathematical Modeling:** formal definition of steps, parameters, performance function, and overall optimization function.

ii. **Computational Implementation:** use of HTML5, CSS3 and JavaScript ES6, with open-source libraries, available on the author's GitHub for academic purposes.:

- **Chart.js**: Pivot Charts and Dashboards
- **jsPDF:** PDF Executive Report Generation
- **SheetJS**: XLSX spreadsheet export
- **MathJax**: LaTeX rendering of formal equations

iii. **Development Process:** structured code on GitHub as an open academic project, allowing for replicability and collaboration.

iv. **Data Entry:** estimated and actual times, operating costs, delivery information, and reverse logistics.

v. **Analysis and Optimization:** identification of bottlenecks, calculation of performance by stage and optimization of α and β weights via genetic algorithm.

vi. **Decision Making:** Dashboards provide intelligent alerts and interpretable reports for strategic and operational decisions.

The framework allows you to simulate scenarios, test parameter adjustments, and evaluate impacts on global supply chain performance, promoting evidence-based decisions.

5. Discussion

SCOPE-IA 4.0 demonstrates that the mathematical formalization of SCM allows integration between predictive analysis, multicriteria evaluation, and heuristic optimization. The approach provides transparency in decision**,** auditability, and support for operational resilience, aligning with Supply Chain 4.0 principles.

The framework offers advantages such as reduced operational uncertainty, increased overall efficiency, early detection of bottlenecks, and the ability to perform trade-offs between

time and cost. In addition, its implementation in web technologies and open-source libraries makes the project replicable and scalable, allowing academic application and future industrial validation.

The theoretical support of Lambert, Cooper & Pagh (1998), Christopher (2016), Sheffi (2005) and Choi, Chan & Yue (2019) underlines the need for integration, visibility, resilience and use of artificial intelligence in SCM, validating the SCOPE-IA 4.0 approach as a significant academic contribution.

6. Conclusion

SCOPE-IA 4.0 presented an academic, formal, and multi-criteria framework for predictive supply chain optimization, meeting the proposed objectives of integrating predictive analytics, multi-criteria evaluation, and heuristic optimization into a replicable and auditable system. The model demonstrated the ability to identify operational bottlenecks, optimize resource allocation, improve overall efficiency, and reduce uncertainties, offering robust support for strategic and operational decision-making in Supply Chain 4.0 environments.

As an academic, open, and non-profit project, SCOPE-IA 4.0 contributes significantly to the area of research and development in Supply Chain Management (SCM), providing a methodological resource for future studies, teaching, and experimentation. The availability of the source code on GitHub allows replicability, collaboration, and extension of the model by other researchers, strengthening the integration between theory and academic practice.

Among the limitations identified are the dependence on reliable input data, the simplicity of compensatory linear assumptions in the performance function, and the need for empirical validation in real industry scenarios, aspects that may impact the generalization of the results.

As suggestions for future work, it is recommended:

- Validation with real industry datasets, to evaluate the effectiveness of the model in practical contexts.
- Integration of deep learning techniques for more accurate demand forecasting.
- Expansion to non-linear multiobjective optimization, allowing simultaneous consideration of multiple complex criteria.

- Incorporation of sustainability and environmental impact metrics, aligning the framework with the contemporary challenges of responsible supply chain management.

In this way, SCOPE-IA 4.0 represents not only a relevant methodological contribution, but also a starting point for future research that seeks to deepen the integration between artificial intelligence, predictive analytics, and systemic optimization in SCM.

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